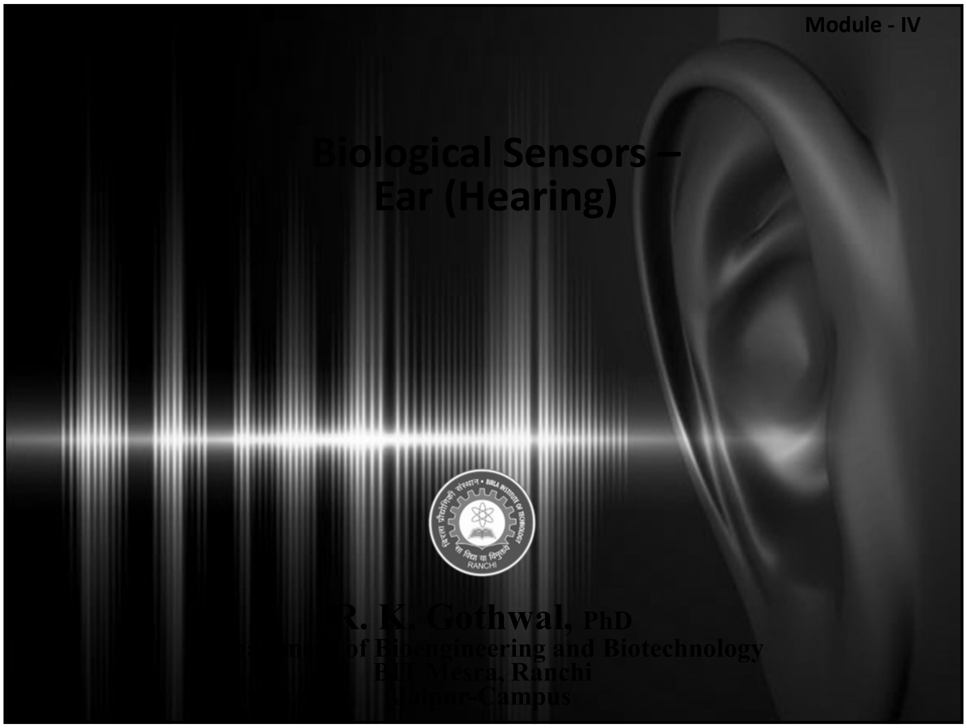

Module - IV

Biological Sensors – Ear (Hearing)

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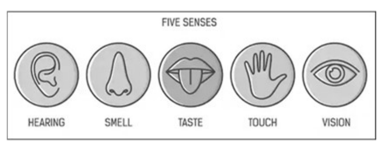
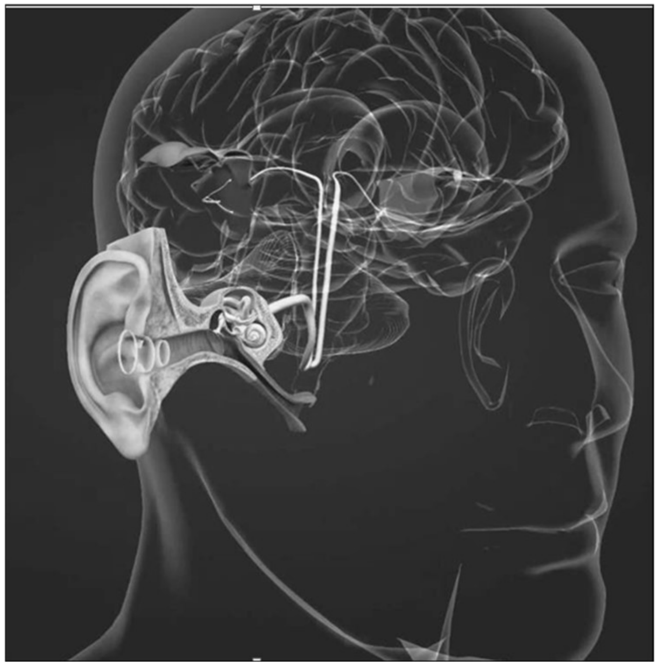
Brain: The human microcontroller

- Brain in human body controls various physiological and mechanical activities through nervous system.
- Humans have five basic **senses**: sight, hearing, smell, taste and touch.
- Living organisms make use of their senses to acquire information from outside environment and communicate this information to the brain to take action.
- The sensory organs transform these signals into electro-chemical message and convey these messages to brain in the form of nerve impulse (action potential).
  – Sensory Nerve cell (neuron) sends impulse to the brain.
  – Motor neuron brings back the triggered action from the brain (what to do).



Hearing



Hearing

- **Hearing** is a process by which the ear transforms sound vibrations in the external environment into **nerve impulses** that are conveyed to the brain, where they are interpreted as sounds.
- Hearing, or **auditory perception**, is the ability to perceive sounds by detecting vibrations, changes in the pressure of the surrounding medium through time, through an organ such as the ear.
- The academic field concerned with hearing is **Auditory Science**.
- Partial or total inability to hear is called **hearing loss**.
- In humans and other vertebrates, hearing is performed primarily by the auditory system: mechanical waves, known as vibrations, are detected by the ear and transduced into **nerve impulses** that are perceived by the brain (primarily in the **Temporal Lobe**).

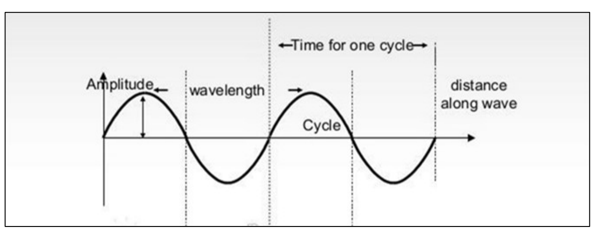
Sounds

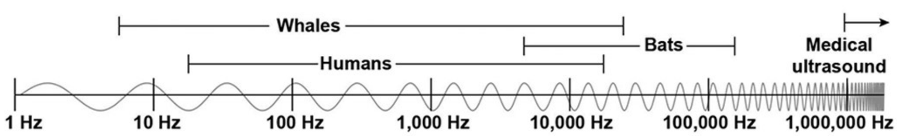
- Sounds are produced when vibrating objects, such as the plucked string of a guitar, produce pressure pulses of vibrating air molecules, better known as sound waves.
- **Sound waves** are longitudinal **waves** that travel through a medium like air or water. When we think about **sound**, we often think about how loud it is (intensity) and its pitch (frequency).
- The ear can distinguish different subjective aspects of a sound, such as its loudness and **pitch**, by detecting and analyzing different physical characteristics of the waves.
- Pitch is the perception of the **frequency** of sound waves i.e., the number of wavelengths that pass a fixed point in a unit of time.
- Frequency is usually measured in cycles per second, or hertz.
- The human ear is most sensitive to and most easily detects frequencies of **1,000 to 4,000 Hz**, but at least for normal young ears the entire audible range of sounds extends from about 20 to 20,000 Hz.
- Sound waves of still higher frequency are referred to as ultrasonic, although they can be heard by other mammals.

Characteristics of Sound Waves

Compression

Rarefaction

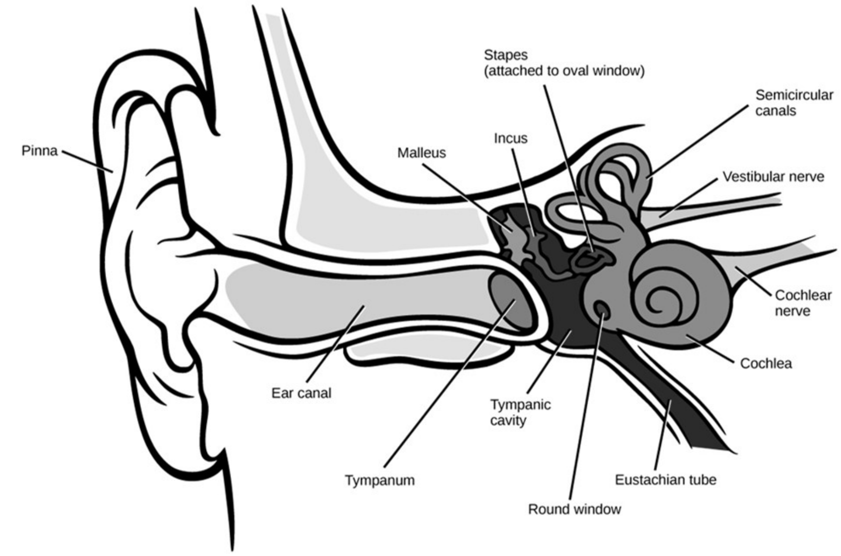




Frequencies of sound and average range of hearing

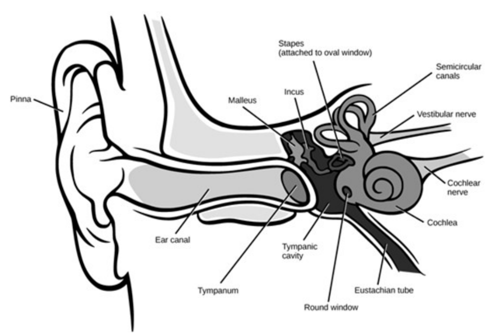
Long wavelength	←	**Short wavelength**
Low frequency	→	**High frequency**
Low energy	→	**High energy**

Hearing System

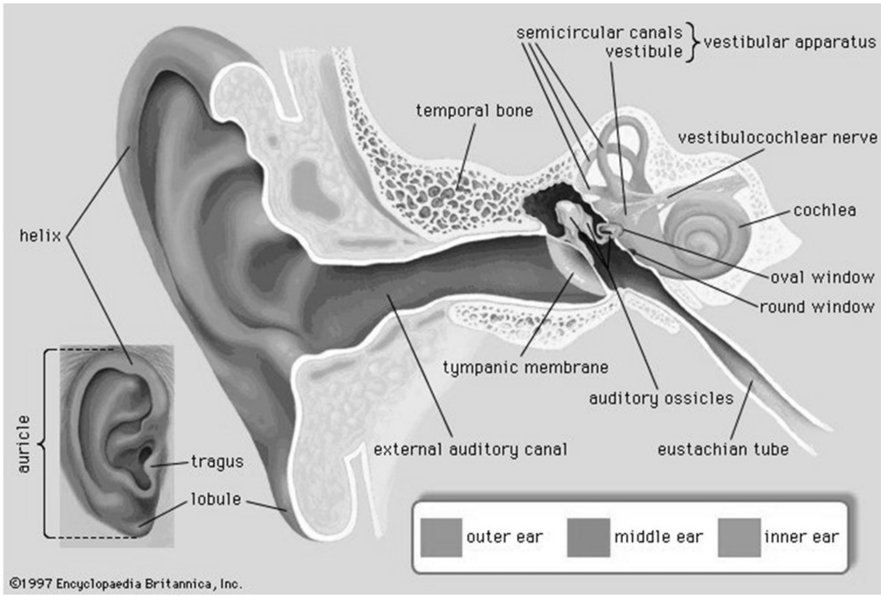


Hearing System

- There are three main components of the human auditory system:
  - the outer ear,
  - the middle ear, and
  - the inner ear.

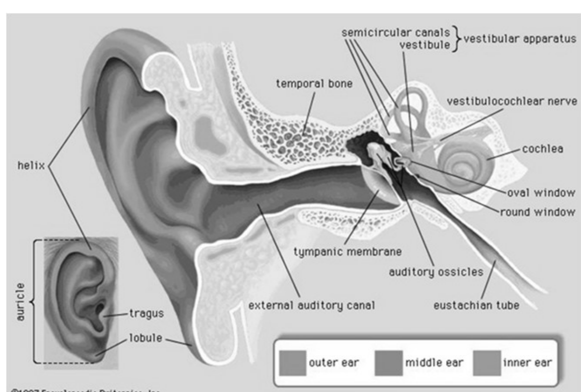


Structure of the human ear



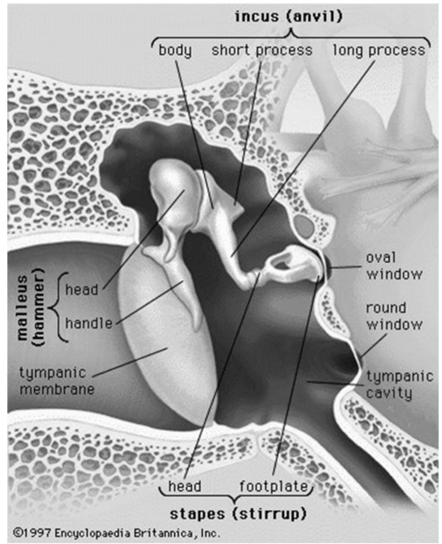
Outer Ear

- The outer ear includes the **pinna**, the visible part of the ear, as well as the ear canal which terminates at the **eardrum**, also called the **tympanic membrane**.
- The pinna serves to focus sound waves through the ear canal toward the eardrum.
- The eardrum is an airtight membrane, and when sound waves arrive there, they cause it to vibrate following the **waveform** of the sound.



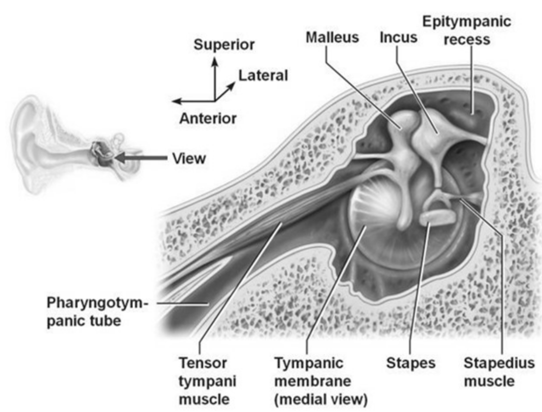
Middle Ear

- The middle ear consists of a small air-filled chamber that is located medial to the eardrum.
- Within this chamber are the three smallest bones in the body, known collectively as the **ossicles** which include the malleus, incus, and stapes (also known as the hammer, anvil, and stirrup, respectively).
- They aid in the transmission of the vibrations from the eardrum into the inner ear, the **cochlea**.
- The purpose of the middle ear ossicles is to overcome the impedance mismatch between air waves and cochlear waves, by providing impedance matching.



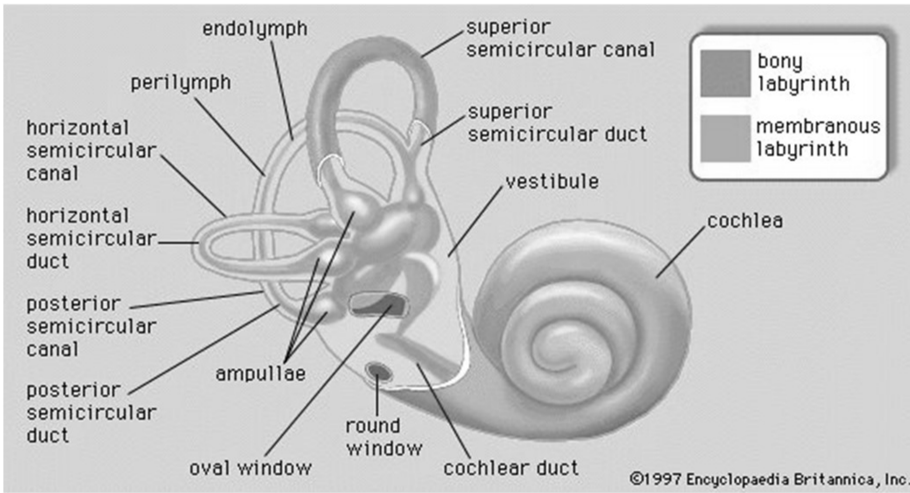
Middle Ear

- Also located in the middle ear are the **stapedius muscle** and **tensor tympani muscle**, which protect the hearing mechanism through a stiffening reflex.
- The **stapedius** is the smallest skeletal muscle in the human body. At just over one millimeter in length, its purpose is to stabilize the smallest bone in the body, the stapes.
- The stapes transmits sound waves to the inner ear through the **oval window**, a flexible membrane separating the air-filled middle ear from the fluid-filled inner ear.
- The **tensor tympani** is a muscle within the ear, located in the bony canal above the osseous portion of the auditory tube. Its role is to damp loud sounds, such as those produced from chewing, shouting, or thunder. Because its reaction time is not fast enough, the muscle cannot protect against hearing damage caused by sudden loud sounds, like explosions or gunshots.
- The **round window**, another flexible membrane, allows for the smooth displacement of the inner ear fluid caused by the entering sound waves.



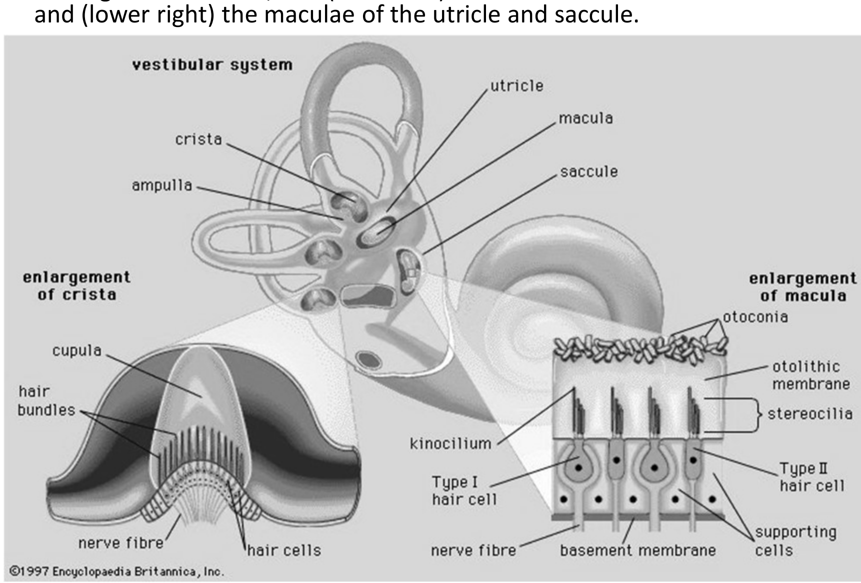
Inner Ear

- The inner ear consists of the **cochlea**, which is a spiral-shaped, fluid-filled tube.

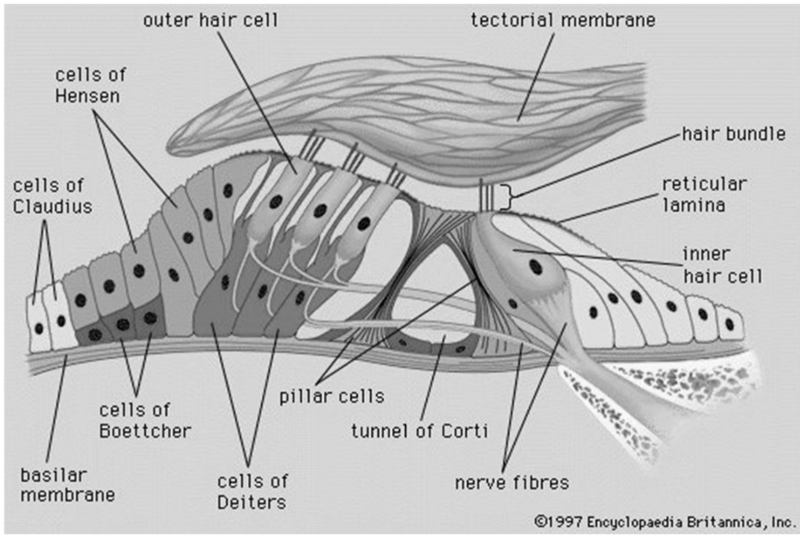


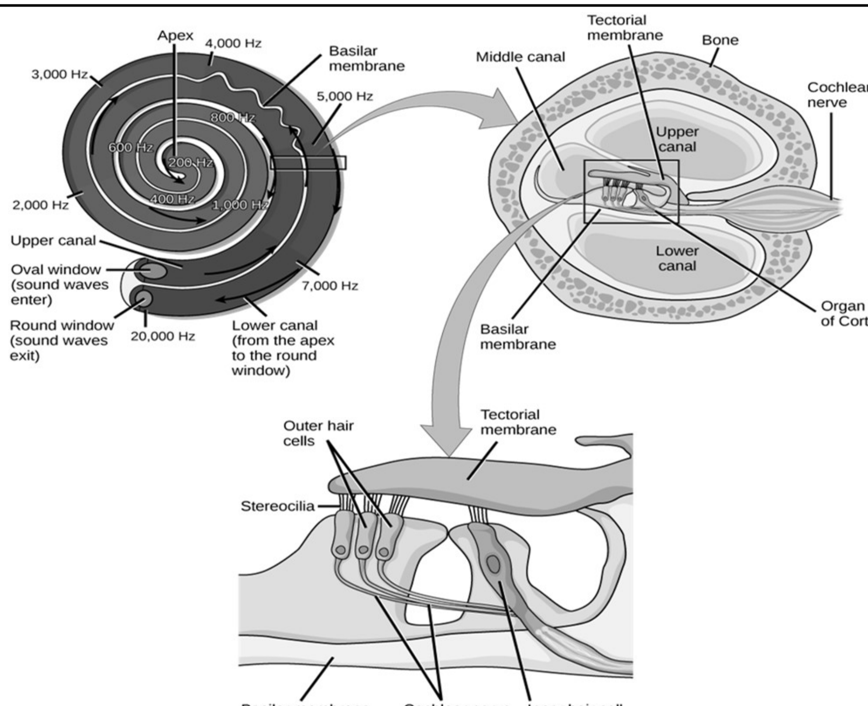
Inner Ear

- The membranous labyrinth of the vestibular system (centre), which contains the organs of balance, and (lower left) the cristae of the semicircular ducts and (lower right) the maculae of the utricle and saccule.

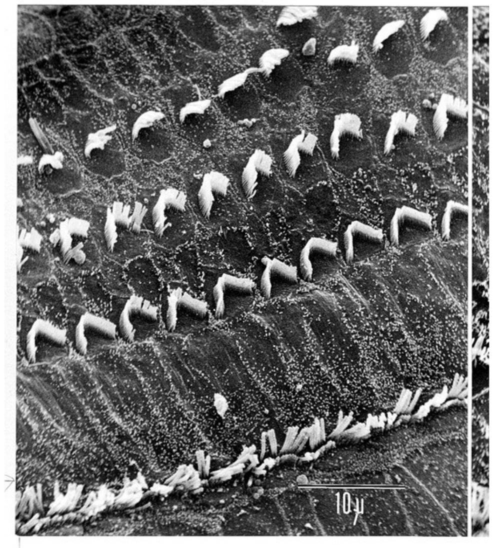


Structure of the organ of Corti





Healthy organ of Corti from a guinea pig

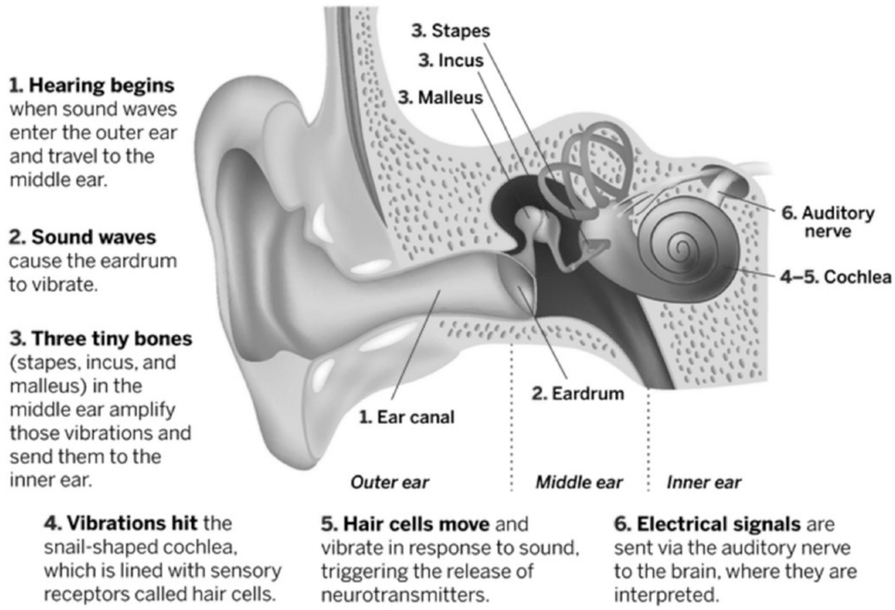


Portion of a healthy organ of Corti from a guinea pig showing the characteristic three rows of outer hair cells and single row of inner hair cells.

Damaged organ of Corti from a guinea pig

Portion of a noise-damaged organ of Corti from a guinea pig exposed to sound at a 120-decibel level, similar to that experienced at a heavy metal rock concert, showing "scars" that have replaced many of the outer hair cells and showing the remaining stereocilia in disarray. Hearing is permanently damaged because lost hair cells will not be replaced, and injured cells may be dying.

Mechanism of Hearing

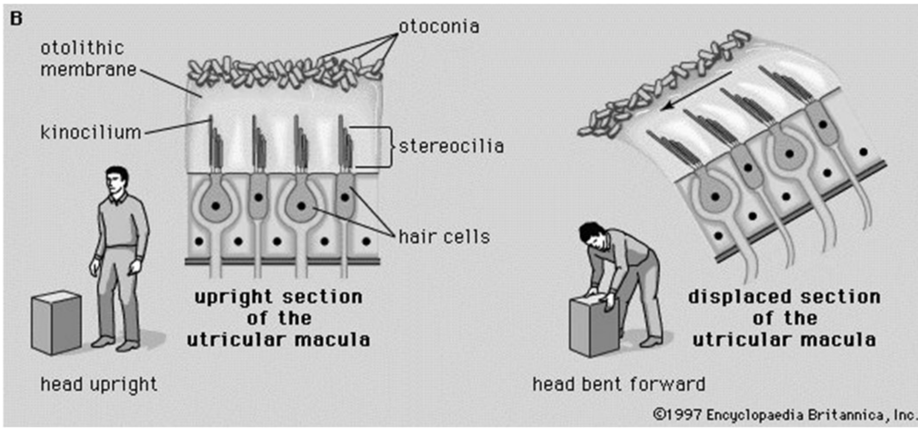


Mechanism of Hearing

- Sound waves enter the outer ear and travel through the external auditory canal until they reach the tympanic membrane, causing the membrane and the attached chain of auditory ossicles to vibrate.
- The mallus will receive the sound and amplify the vibrations across the ossicles in the middle ear.
- The osicles push the sound waves with greater vibrations to the oval window.
- These vibrations in the oval window will generate fluid within the vestibular duct to move the waves into cochlear duct.
- These waves cause tectorial membrane to bounce and also movement of basilar membrane cause hair cells to bend.
- When the hair cells bend, ion channel opens and release neurotransmitters into the synapse with the sensory neuron, causing it to change the rate of Action Potential firing to the brain via cochlear nerve.
- The cochlear nerve carries the impulses to primary auditory neurons in the medulla.
- The information will reach to the secondary neurons in the mid brain and thalamus before projecting into the auditory cortex portion of the brain.

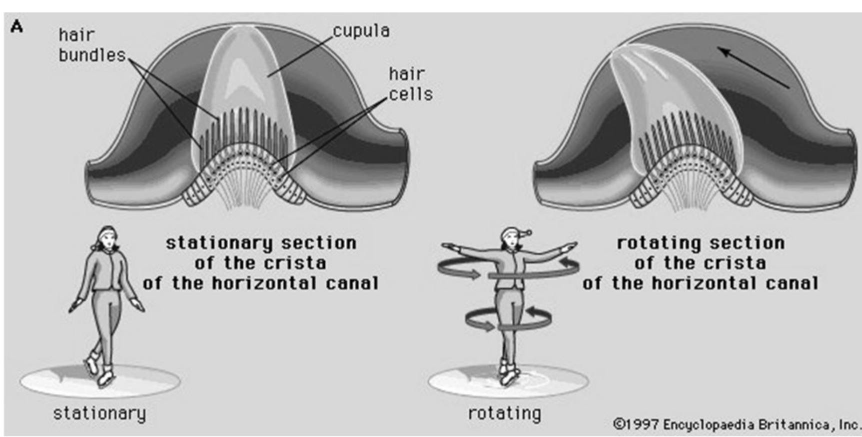
How ear balances your body

- In vertebrates the utricular maculae in the inner ear contain an otolithic membrane and otoconia (particles of calcium carbonate) that bend hair cells in the direction of gravity. This response to gravitational pull helps animals maintain their sense of balance.



How ear balances your body

- The cristae of the semicircular ducts, which form one of the two sensory organs of balance (the second being the maculae of the utricle and saccule), respond to rotational movements and are involved in dynamic equilibrium.



Thanks